Collecting NAND dumps

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Before a NAND dump file can be analyzed, someone somewhere have to have collected the file from a unit where the problem originated. That someone can be a tester, customer, factory engineer, etc. So they have to have the right tools and instructions to be able to collect the files. The main tool to collect the files is via fastboot tool. There are 2 situations that will dictate how you approach the collection of the files:

1. System boots partially and operator of the unit can send it to fastboot mode
2. System either doesn't boot at all or it boots into a mode where fastboot is not possible

Using built-in fastboot mode

If the unit partially boots and it can be sent to Fastboot mode using

reboot bootloader from serial port or **adb shell reboot bootloader** from CMD

then you are lucky and all you will need to do is to run the following command from Windows while in fastboot mode:

fastboot dump all.raw <filename>

The **.raw** suffix is used to collect in addition to the data bytes within the NAND pages to also collect what is called out-of-band or spare bytes. The out-of-band or spare-bytes are used for ECC error correction and they are also used as part of the strategy to mark bad blocks. So when you use .raw as an extension, you are doing a faithful copy of the flash capturing bad block information and ECC information.

Things to know about collecting files with the .raw option

- If you open the file with an hexadecimal editor, you can't modify bytes of data and write them back to a unit and expect them to be recognized by the system. Since the data is corrected by ECC, the values you modify will be taken as bitflips or errors. To properly change data in a file collected with .raw option, you would have to recompute the FEC code for the sub-page where you are making the modification. This is too complex to do it in practice.
- There could be bening bitflips in the data so you could potentially see what appears to be incorrect data but it could just be the result of a bitflip and it could be normal
- With the .raw option, even bad blocks will be collected and those can be identified visually with an hexadecimal editor
- EFS2 can only be collected with .raw option since this file system type uses the out-of-band area for file system operation. So if you are transplanting the EFS2 of one unit to another, you would have to dump it using .raw option and also flashing it with the same option.
- If you dump the entire NAND contents and if the part has 512MB of data capacity, the resultant file won't be 524,288 KB long. It will be 540,672 KB long. That is because it also has the out-of-band bytes included.
- For a 512MB part, there will be
    - 4096 blocks
    - 64 pages per block
    - each block has 132KB (128KB data, 4KB spare)
    - each page has 2112 bytes (2048 data, 64 spare)
    - total capacity is 540,672KB (512MB data, 16MB spare)
- If you dump the NAND contents using the .raw option, then you have to flash it back also with .raw option

Using Qualcomm tools

Dumping of NAND contents on SA515M HW is done with the assistance of QFIL tool. The unit has to be place first in Emergency Download (EDL).

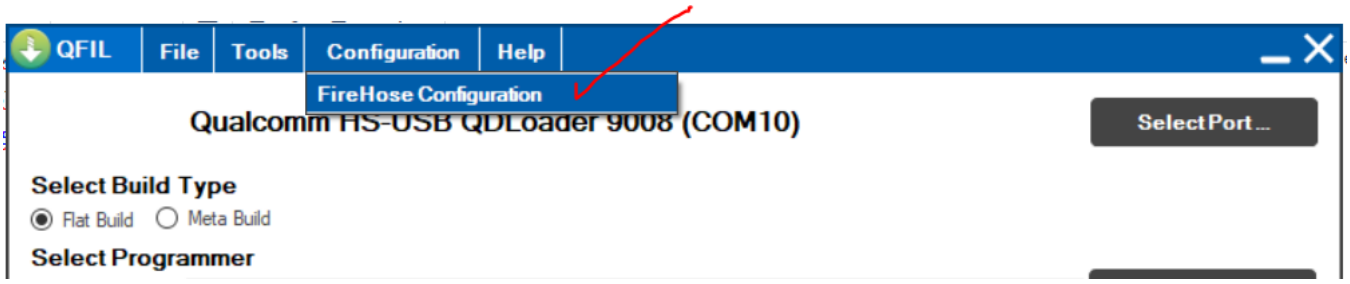
QFIL is part of the suite of tools provided by QPST. If you have QPST installed in your system, you already have QFIL. In addition to QFIL, you will need the following 2 files:

1. prog_firehose_sdx24.mbn for SA415M (or prog_firehose_sdx55.mbn for SA515M) which should be available with the bundle of images required for flashing units.
2. **dump_nand_contents.xml**

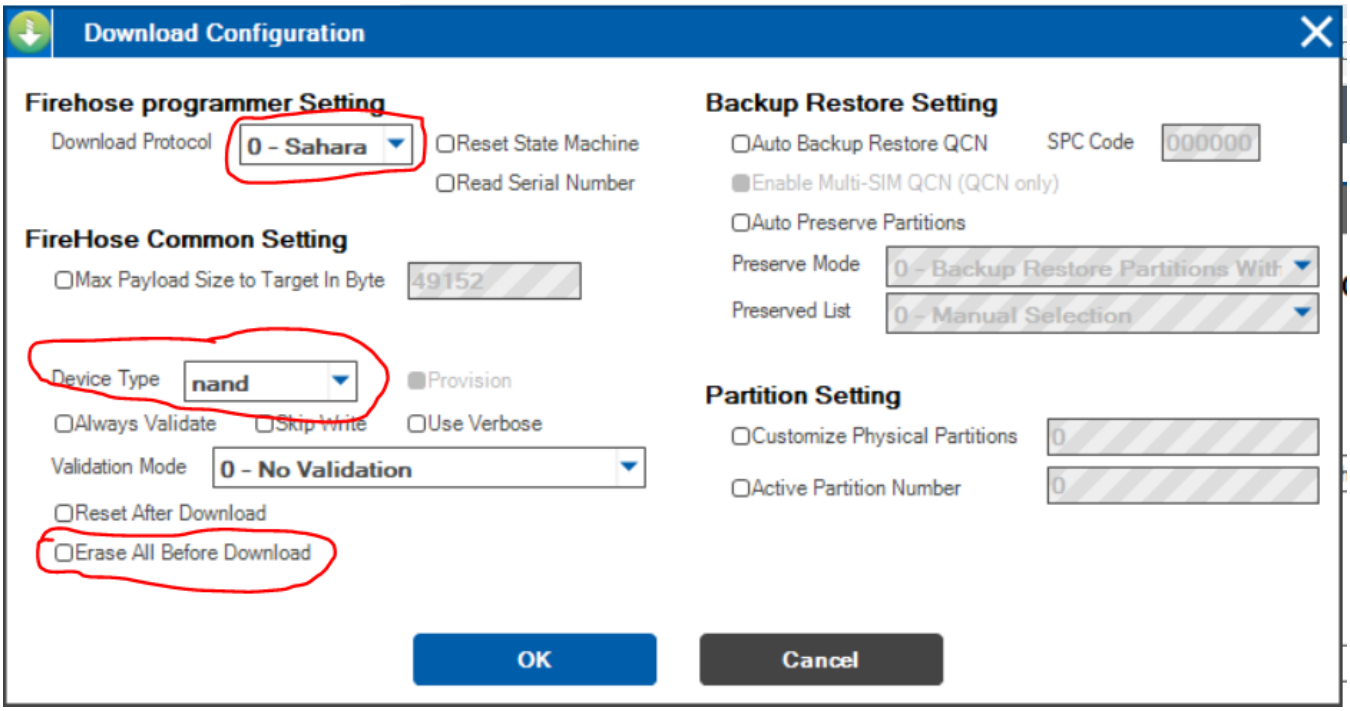
Please open the XML file for additional instructions.

Configuring QFIL for dumping NAND contents

Select FireHose Configurtion as shown:



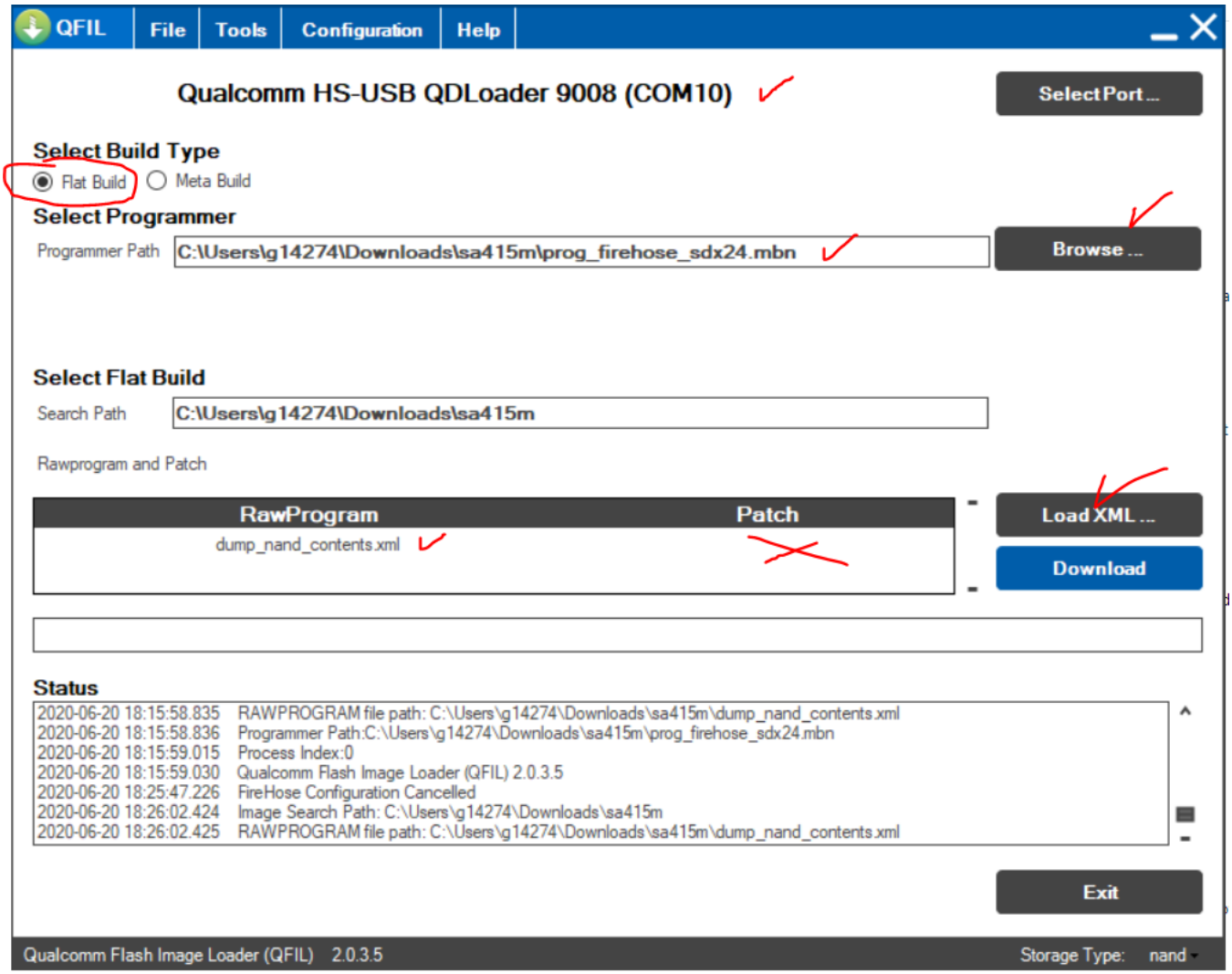
Please make sure the configuration is set as follows for highlighted items:



It is **extremely important** that "Erase All Before Download" option is unchecked or otherwise you will erase the flash and the dump NAND contents will be blank.

Performing NAND dump

With the 2 files mentioned earlier available, open QFIL and select those 2 files by pressing LOAD XML button:



When ask for patch file after pressing "Load XML" and selecting "dump_nand_contents.xml" as "RawProgram", press cancel when prompted from patch file. The "Patch" field above should be empty. Please sure "Storage Type" is set as "nand" and that "Select Build Type" es set as "Flat Build".

Once is everything set as indicated, select "Download". It will take more than 10 minutes to dump the NAND contents so be patient. The resultant output will have the name "all.bin" and it will be located in this path:

C:\Users\<user id>\AppData\Roaming\Qualcomm\QFIL\COMPORT_##

In the above example, ## is 10 (i.e. COM10).